Topic

Library Book Search System Using BST with Insertion, Searching, Deletion and Sorted Listing

Problem Statement

Develop a C++ program to manage library book records using a Binary Search Tree (BST). Each record should contain a unique book identification number and its title. The program should support insertion of new books, searching for books by ID, deletion of book records, and displaying all books in ascending order of their IDs. It should also count and display the number of comparisons performed during each search operation. The BST property must be maintained, and duplicate book IDs should not be allowed.

Algorithm

1. Start.
2. Initialize the BST root as NULL.
3. Display the menu with options to insert, search, delete, display sorted records, and exit.
4. Read the user's choice.
5. If the choice is **Insert Book**:
    - Read the book ID and title.
    - Check whether the ID already exists.
    - If it exists, display a duplicate ID message.
    - Otherwise, create a new node and insert it into the correct BST position.
    - Display a success message.
6. If the choice is **Search Book**:
    - Read the book ID.

- Initialize the comparison counter to zero.
- Compare the ID with each visited node and increment the counter.
- If the ID matches, display the book ID and title.
- If the ID is smaller, search the left subtree; otherwise, search the right subtree.
- If the ID is not found, display "Book not found."
- Display the total number of comparisons.

7. If the choice is **Delete Book**:
    - Read the book ID.
    - Search for the ID in the BST.
    - If it is not found, display a message.
    - If the node has no children, delete it directly.
    - If the node has one child, replace it with its child.
    - If the node has two children, replace it with its inorder successor and delete the successor node.
    - Display the result.
8. If the choice is **Display Books in Sorted Order**:
    - Perform inorder traversal of the BST.
    - Visit the left subtree, display the current book ID and title, and then visit the right subtree.
    - Display all books in ascending order of their IDs.
9. If the choice is **Exit**, terminate the program.
10. If the choice is invalid, display an error message.
11. Repeat the menu until the user chooses Exit.
12. Stop.